FoDS TDA FEATURES EXPLANATIONS

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1 Topological Feature Set

This section provides a comprehensive description of the topological features extracted from persistence diagrams obtained through sliding-window embeddings of time series. The features are organized into nine major families: persistence entropy, amplitude measures, Betti curve statistics, persistence landscapes, silhouette functions, heat kernel signatures, raw diagram statistics, Carlsson coordinates, and cross-dimensional ratios. Unless otherwise stated, features are computed separately for homology dimensions H_0 and H_1.

1.1 Persistence Entropy

Features: `entropy_0`, `entropy_1`

Given a persistence diagram with lifetimes $l_i = d_i - b_i$ and normalized weights

$$p_i = \frac{l_i}{\sum_j l_j},$$

the *persistence entropy* is defined as

$$\text{Entropy} = -\sum_i p_i \log(p_i).$$

Higher entropy values indicate more uniformly distributed feature lifetimes, whereas lower entropy indicates dominance by a few long-lived features.

1.2 Amplitude Features

Features:

- Bottleneck: `amplitude_bottleneck_0`, `amplitude_bottleneck_1`
- Wasserstein: `amplitude_wasserstein_0`, `amplitude_wasserstein_1`
- Landscape norm: `amplitude_landscape_0`, `amplitude_landscape_1`

Let D be a diagram and Δ the diagonal. Amplitude features measure distances between D and Δ:

$$\text{Bottleneck}(D, \Delta) = \inf_{\gamma} \sup_{x \in D} \|x - \gamma(x)\|_\infty,$$

$$W_p(D, \Delta) = \left(\inf_{\gamma} \sum_{x \in D} \|x - \gamma(x)\|^p \right)^{1/p},$$

$$\|\lambda\|_p = \left(\int |\lambda(t)|^p \, dt \right)^{1/p},$$

where $\lambda(t)$ denotes the persistence landscape.

1.3 Betti Curve Statistics

Features: `betti_L1_*`, `betti_L2_*`, `betti_max_*`, `betti_mean_*`, `betti_median_*`, `betti_std_*`, `betti_var_*`, `betti_skew_*`, `betti_kurt_*`, `betti_AUC_*`, `betti_gini_*`, `betti_nonzero_count_*`, `betti_sparsity_*`

A Betti curve is defined as

$$\beta_k(t) = \#\{\text{features alive at scale } t\}.$$

For each Betti curve, the following statistics are computed:

$$\|\beta\|_1 = \sum_t |\beta(t)|, \quad \|\beta\|_2 = \left(\sum_t \beta(t)^2 \right)^{1/2},$$

as well as max, mean, variance, skewness, kurtosis, the area under the curve (AUC), Gini coefficient, sparsity, and nonzero count.

1.4 Persistence Landscape Statistics

Features: `landscape_L1_*`, `landscape_L2_*`, `landscape_max_*`, `landscape_mean_*`, `landscape_median_*`, `landscape_std_*`, `landscape_var_*`, `landscape_skew_*`, `landscape_kurt_*`, `landscape_AUC_*`, `landscape_gini_*`, `landscape_sparsity_*`

A persistence landscape $\lambda_k(t)$ is a stable functional representation of the diagram, and the same statistical family as above is applied to each landscape.

1.5 Silhouette Statistics

Features: `silhouette_L1_*`, `silhouette_L2_*`, `silhouette_max_*`, `silhouette_std_*`, `silhouette_AUC_*`, `silhouette_gini_*`, `silhouette_sparsity_*`, etc.

The silhouette function is defined by

$$s_k(t) = \frac{\sum_i l_i^\alpha \Lambda_i(t)}{\sum_i l_i^\alpha},$$

where $\Lambda_i(t)$ are landscape functions. Features are computed using the same statistical measures as for Betti curves and landscapes.

1.6 Heat Kernel Features

Features: `HK_L1_*`, `HK_L2_*`, `HK_mean_*`, `HK_std_*`, `HK_energy_*`, `HK_entropy_*`, `HK_sparsity_ratio_*`, `HK_row_L1_mean_*`, `HK_col_L1_std_*`, `HK_SV1_*`, `HK_SV2_*`, `HK_SV3_*`, `HK_Laplacian_eig_min_*`, `HK_Laplacian_eig_max_*`, `HK_Laplacian_trace_*`, `HK_Laplacian_fro_*`

Given persistence points x_i, the heat kernel matrix is

$$H(i,j) = \exp\left(-\frac{\|x_i - x_j\|^2}{4t}\right).$$

We compute:

- Global norms: $\|H\|_1$, $\|H\|_2$
- Statistical moments of $H(i,j)$
- Energy $= \sum_{i,j} H(i,j)^2$
- Entropy and sparsity ratios
- Row/column L_1-norm statistics
- Singular values of H
- Laplacian eigenvalues, trace, and Frobenius norm

1.7 Raw Persistence Diagram Statistics

Features: Counts, lifetime sums, max/mean/median/std/variance, coefficient of variation, Gini coefficient, quantiles (Q90, Q95, Q99), skewness, kurtosis, birth/death ranges, birth–death correlation, centers of mass (COM), diagonal distance, and dominance shares (top-1, top-3, top-5).

Let $D = \{(b_i, d_i)\}$ and $l_i = d_i - b_i$. Example statistics:

$$\mathrm{CV} = \frac{\sigma(l_i)}{\mu(l_i)}, \qquad \mathrm{Gini}(l_i) = \frac{\sum_{i,j} |l_i - l_j|}{2N \sum_i l_i}.$$

$$\text{top1\_share} = \frac{\max_i l_i}{\sum_i l_i}.$$

Centers of mass:

$$\mathrm{COM}_b = \frac{1}{N}\sum_i b_i, \qquad \mathrm{COM}_d = \frac{1}{N}\sum_i d_i.$$

Birth–death correlation:

$$\rho = \mathrm{corr}(b_i, d_i).$$

1.8 Carlsson Coordinates

Features: `carlsson_f1`–`carlsson_f5`

Let $D = \{(b_i, d_i)\}$. Carlsson coordinates are defined as:

$$f_1 = \sum_i b_i, \qquad f_2 = \sum_i d_i, \qquad f_3 = \sum_i (d_i - b_i),$$

$$f_4 = \sum_i b_i(d_i - b_i), \qquad f_5 = \sum_i d_i(d_i - b_i).$$

These provide a compact, stable vector representation of persistence diagrams.

1.9 Cross-Dimensional Ratios

Features: `H1_over_H0_sum_lifetimes`, `H1_over_H0_max_lifetime`, `H1_over_H0_betti_AUC`, `entropy_diff_H1_minus_H0`

Example:

$$\frac{\sum_i l_i^{(1)}}{\sum_i l_i^{(0)}}, \qquad \Delta\text{Entropy} = \text{Entropy}_{H_1} - \text{Entropy}_{H_0}.$$

These ratios quantify the relative dominance of H_1 (loop-like) structures compared to H_0 (connectivity).